

29 B

By conservation of charge and mass number, $X = 2$.

$$\text{No. of hydrogen nuclei} = 0.001 / (1.008 \times 1.66 \times 10^{-27}) = 5.98 \times 10^{23}$$

$$\begin{aligned} \text{Energy released, in MeV, in 1 reaction} &= 1.6 \times 10^{12} / (5.98 \times 10^{23} \times 1.6 \times 10^{-13}) \\ &= 16.73 \text{ MeV} \end{aligned}$$

Total $\text{BE}_{\text{He-4}} - \text{BE}_{\text{Li-7}} = \text{energy released, } 16.73 \text{ MeV}$ (note BE of H-1 is 0, as it is only 1 proton by itself)

$$2 \times 28.3 - \text{BE}_{\text{Li-7}} = 16.73$$

$$\text{BE}_{\text{Li-7}} = 39.9 \text{ MeV}$$

30 D